IBM Developer
SKILLS NETWORK

Winning Space Race with Data Science

<Name>
<Date>

Outline

- Executive Summary
- Introduction
- Methodology
- Results
- Conclusion
- Appendix

Executive Summary

- Objective: Predict Falcon 9 first stage landing success
- Methods: Data collection, data wrangling, EDA, SQL, and Machine Learning models
- Models used: Logistic Regression, SVM, Decision Tree, KNN
- Best model: Decision Tree with ~0.89 accuracy
- Key insight: Payload mass and orbit type strongly influence landing success

Introduction

- SpaceX aims to reduce space mission costs through rocket reusability
- Falcon 9 first stage recovery is a key factor in cost efficiency
- Predicting landing success can help optimize mission planning

- Objective: Build a model to predict landing outcome based on launch data

Section 1

Methodology

Methodology

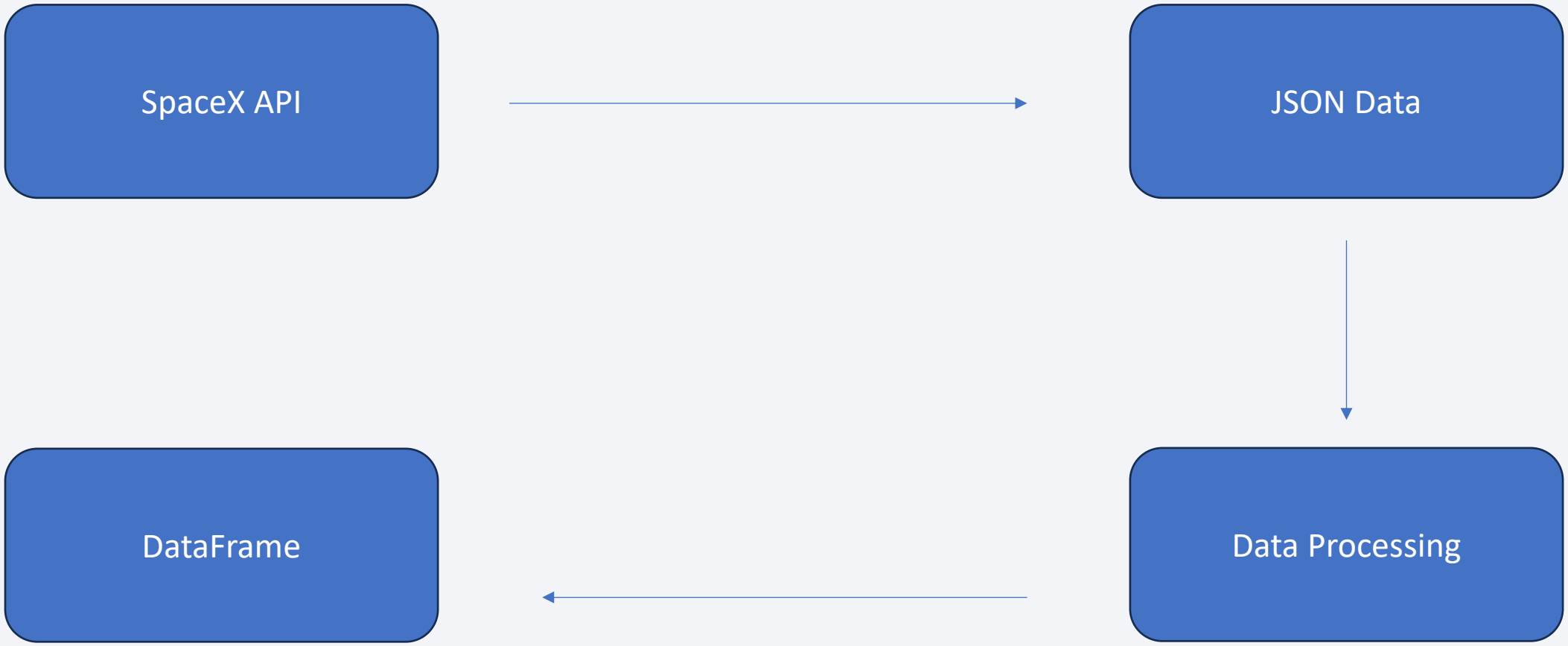
Executive Summary

- Data collection methodology:
  - SpaceX API and Web Scraping from SpaceX Wikipedia page
- Perform data wrangling
  - Cleaning, filtering, and feature selection
- Perform exploratory data analysis (EDA) using visualization and SQL
- Perform interactive visual analytics using Folium and Plotly Dash
- Perform predictive analysis using classification models
  - Model tuning and evaluation using GridSearchCV and cross-validation

Data Collection

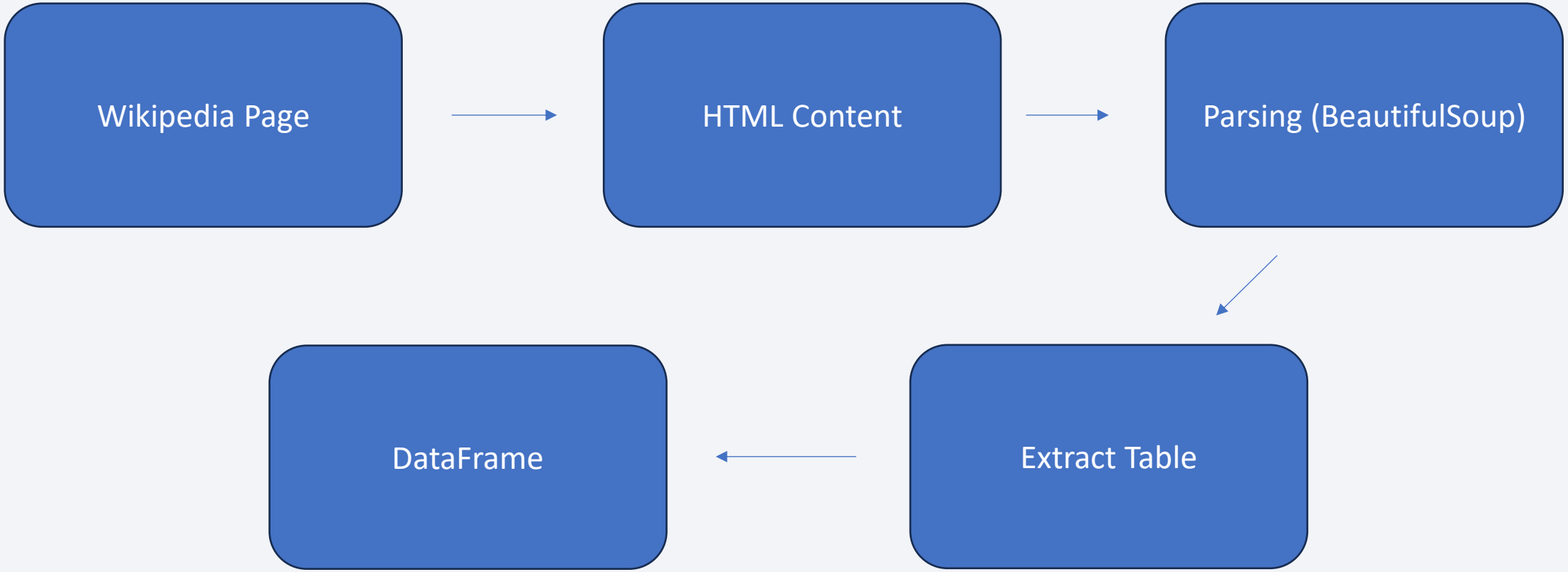
- Data was collected from the SpaceX REST API
- Additional data was obtained through web scraping from Wikipedia
- Key features collected include payload mass, launch site, orbit type, and flight number
- Data was stored and combined into structured datasets for analysis

Data Collection – SpaceX API



https://github.com/yhyl12/SpaceX-Landing-Prediction-Project/blob/main/jupyter-labs-spacex-data-collection-api-v2).ipynb

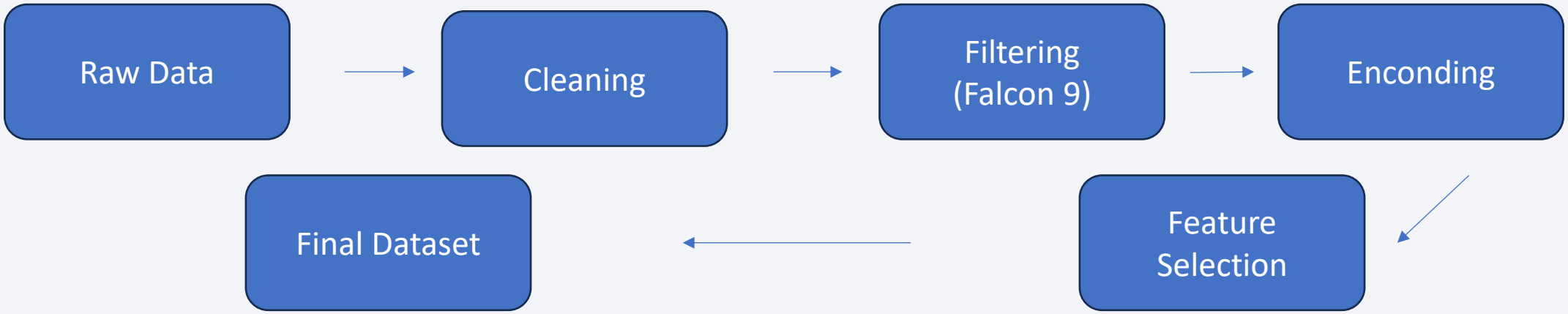
Data Collection - Scraping



https://github.com/yhyl12/SpaceX-Landing-Prediction-Project/blob/main/jupyter-labs-webscraping%20(1).ipynb

Data Wrangling

- Removed missing and irrelevant data
- Filtered launches to include only Falcon 9 missions
- Converted categorical variables into numerical format (encoding)
- Created new features and selected relevant variables
- Final dataset prepared for analysis and modeling



https://github.com/yhyl12/SpaceX-Landing-Prediction-Project/blob/main/labs-jupyter-spacex-Data%20wrangling-v2%20(2).ipynb

EDA with Data Visualization

- Used scatter plots to analyze relationships between variables
- Used bar charts to compare success rates across categories
- Used line charts to analyze trends over time
- Identified key patterns between payload, orbit type, and landing success
- Visualizations helped guide model development

https://github.com/yhyl12/SpaceX-Landing-Prediction-Project/blob/main/edadataviz%20(3).ipynb

EDA with SQL

- Used SQL queries to explore and analyze the dataset
- Retrieved information such as launch sites, payload statistics, and landing outcomes
- Calculated success rates and aggregated metrics
- Identified patterns and trends using structured queries

https://github.com/yhyl12/SpaceX-Landing-Prediction-Project/blob/main/jupyter-labs-eda-sql-coursera_sqllite%20(1).ipynb

Build an Interactive Map with Folium

- Created an interactive map to visualize launch site locations
- Added markers for each launch site
- Used color-coded markers to represent landing outcomes
- Analyzed proximity to geographical features such as coastlines and infrastructure
- Identified patterns related to location and landing success

https://github.com/yhyl12/SpaceX-Landing-Prediction-Project/blob/main/lab_jupyter_launch_site_location.ipynb

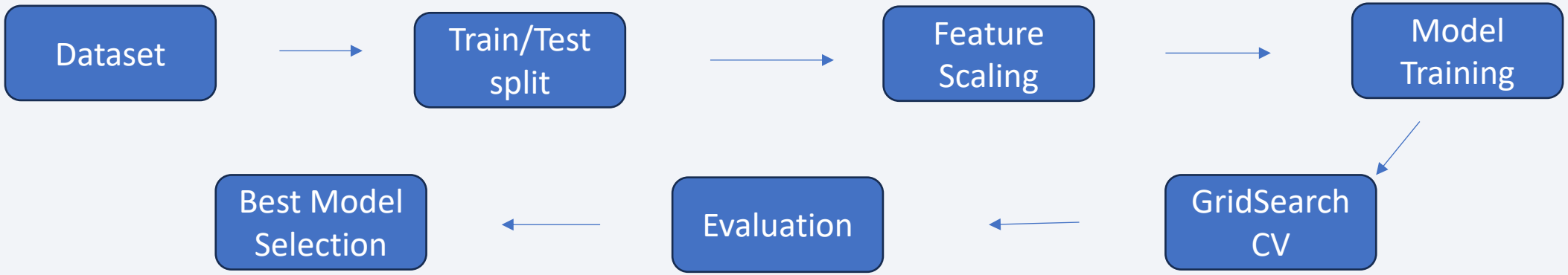
Build a Dashboard with Plotly Dash

- Developed an interactive dashboard using Plotly Dash
- Included pie charts to show launch success distribution
- Added scatter plots to analyze payload vs landing outcome
- Enabled filtering by launch site and payload range
- Allowed dynamic exploration of launch performance

https://github.com/yhyl12/SpaceX-Landing-Prediction-Project/blob/main/spacex-dash-app%20(1)%20(1).py

Predictive Analysis (Classification)

- Built classification models to predict landing success
- Models used: Logistic Regression, SVM, Decision Tree, KNN
- Data was standardized using StandardScaler
- Hyperparameters were optimized using GridSearchCV
- Model performance evaluated using accuracy and confusion matrix



https://github.com/yhyl12/SpaceX-Landing-Prediction-Project/blob/main/SpaceX-Machine-Learning-Prediction-Part-5-v1%20(3).ipynb

Results

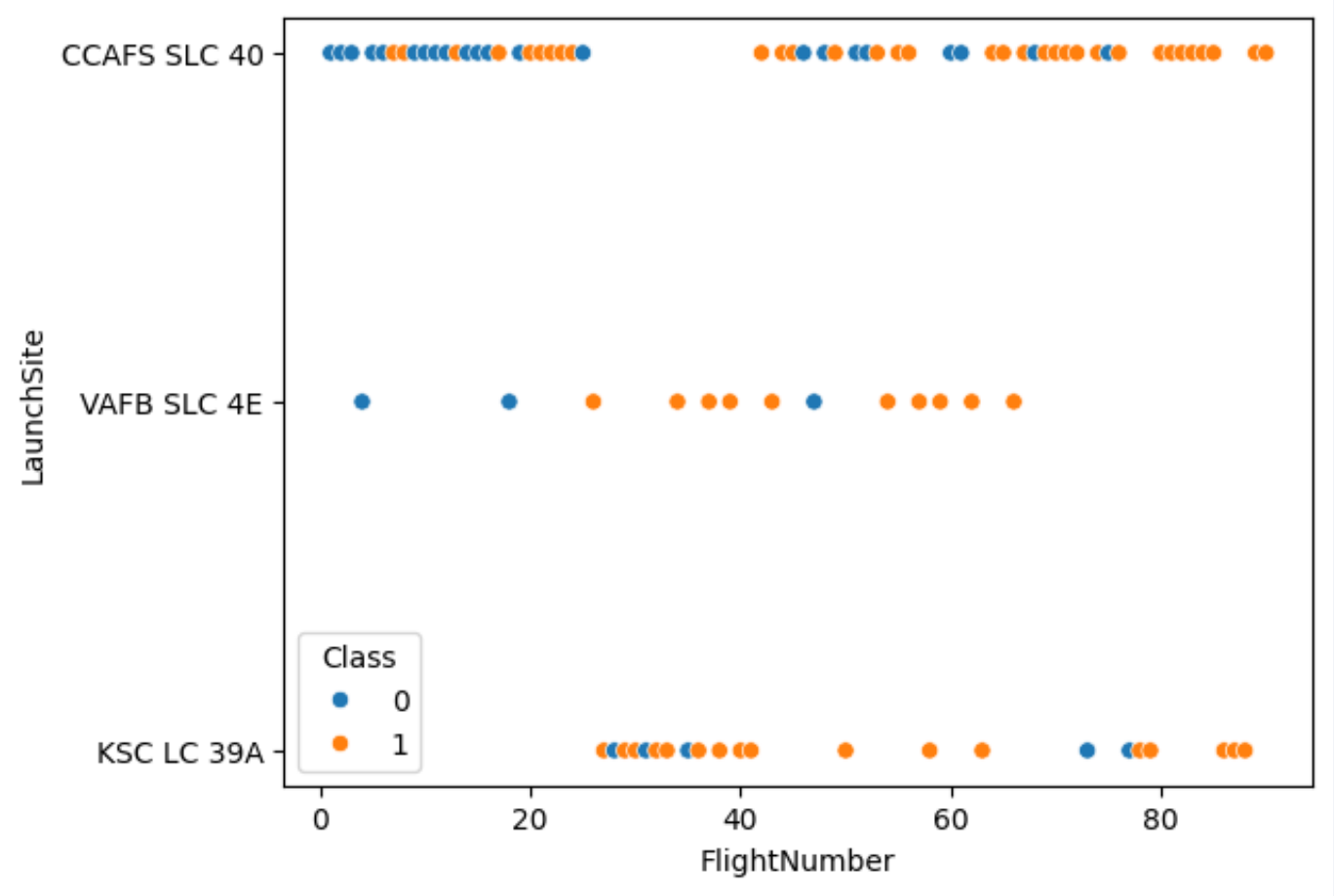
- EDA showed that payload mass and orbit type influence landing success
- Success rate improves over time due to experience and reuse
- Interactive visualizations confirmed patterns across launch sites
- Decision Tree achieved the highest accuracy (~0.89)
- Model predictions show strong performance with few errors

Section 2

Insights drawn from EDA

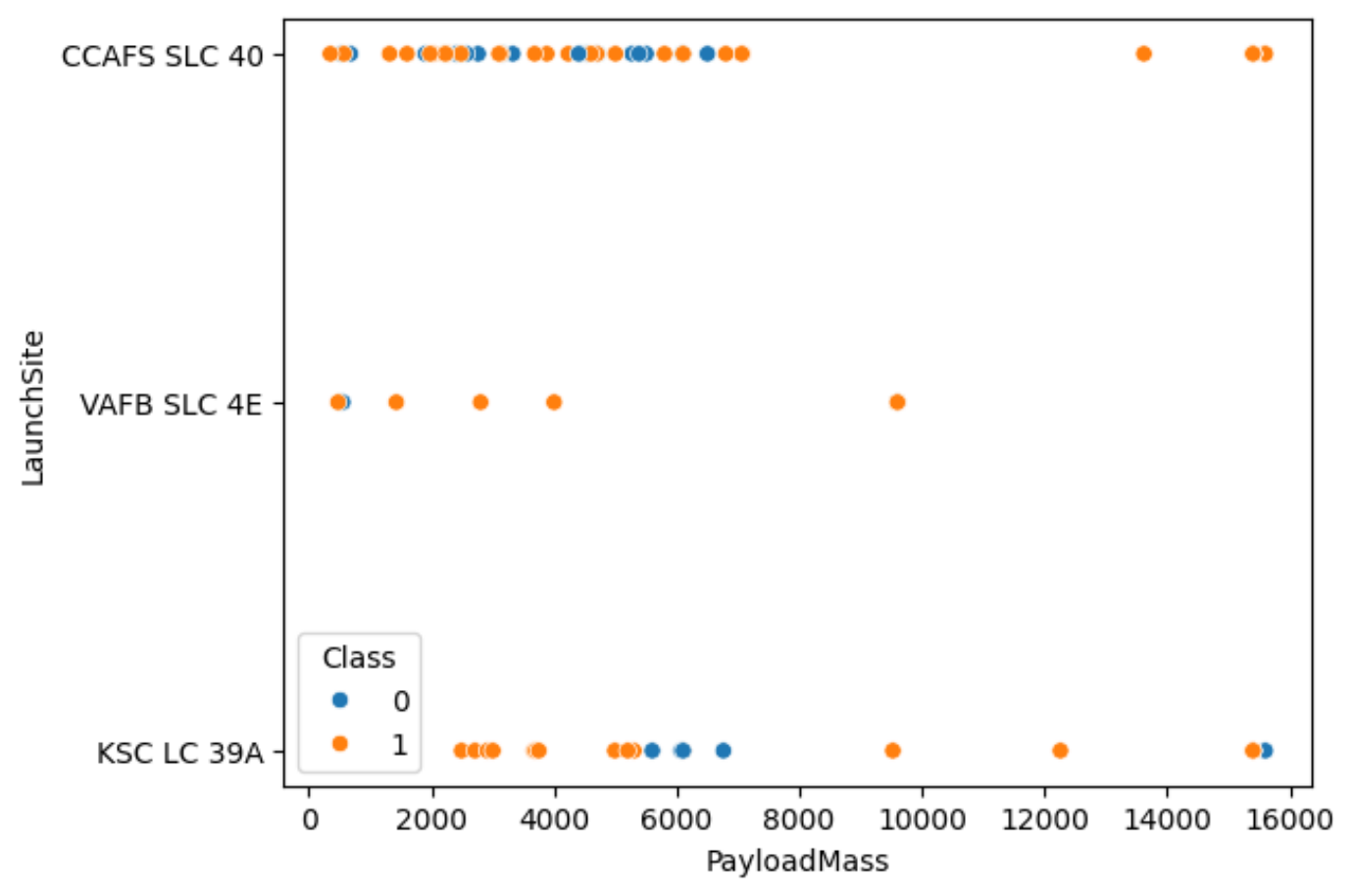
Flight Number vs. Launch Site

- Scatter plot shows flight number distribution across launch sites
- Colors represent landing success (1) and failure (0)
- Successful landings become more frequent at higher flight numbers
- Indicates improvement in landing performance over time



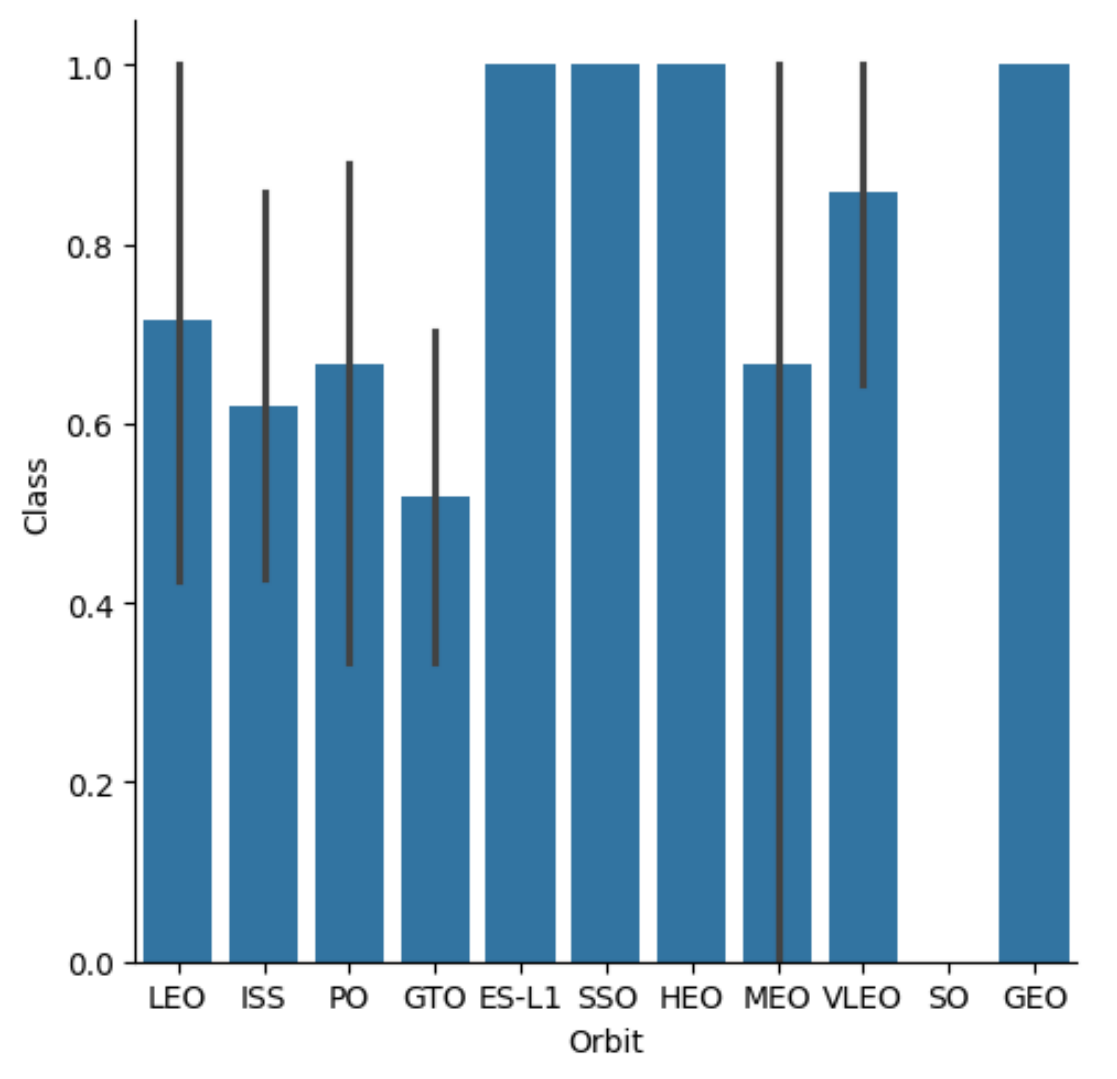
Payload vs. Launch Site

- Scatter plot shows payload mass distribution across launch sites
- Colors represent landing success (1) and failure (0)
- Lower payloads are more likely to result in successful landings
- Higher payloads show more variability in landing outcomes



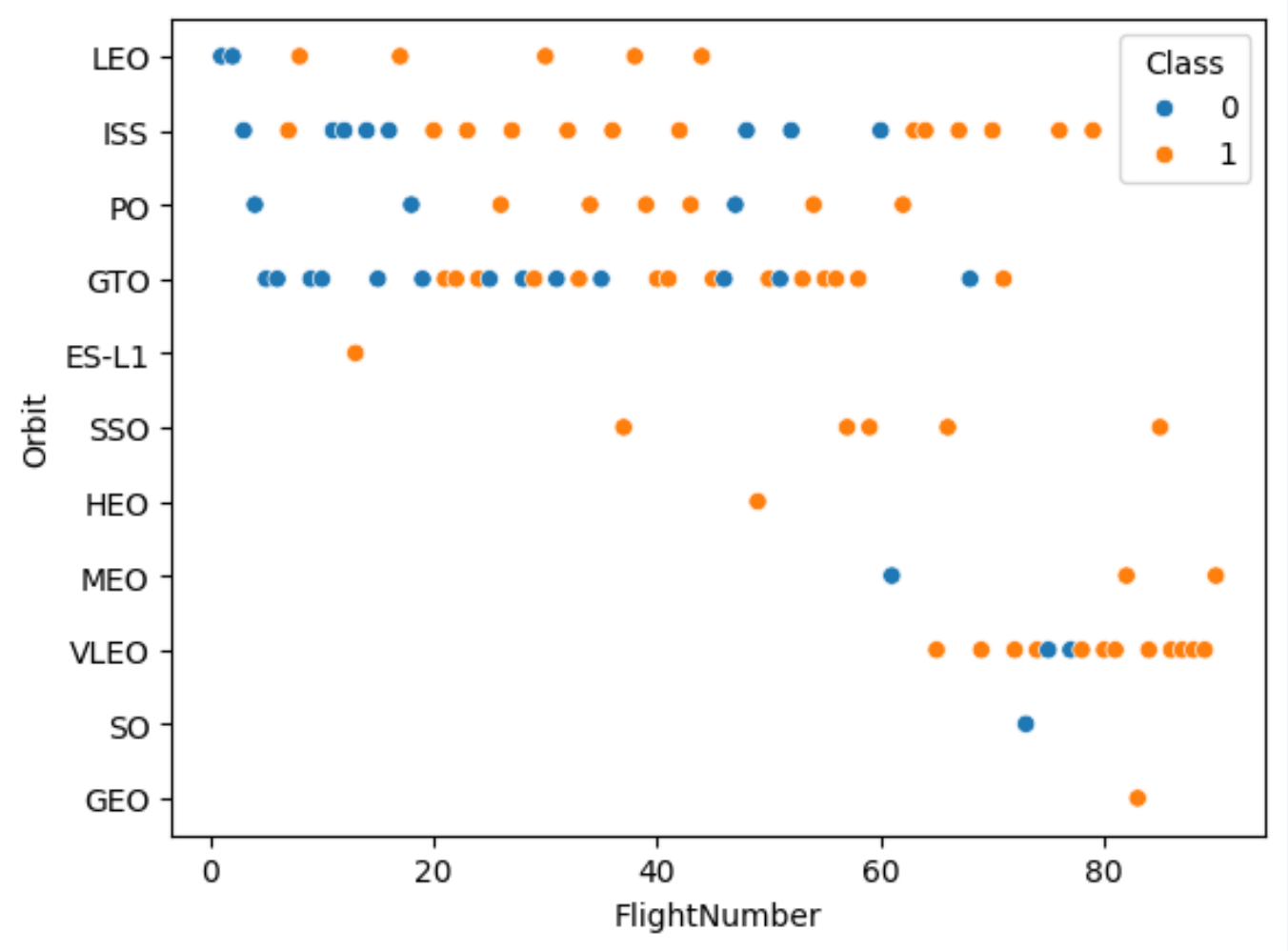
Success Rate vs. Orbit Type

- Bar chart shows landing success rate for each orbit type
- Some orbits have significantly higher success rates
- LEO orbits show higher landing success
- More complex orbits tend to have lower success rates



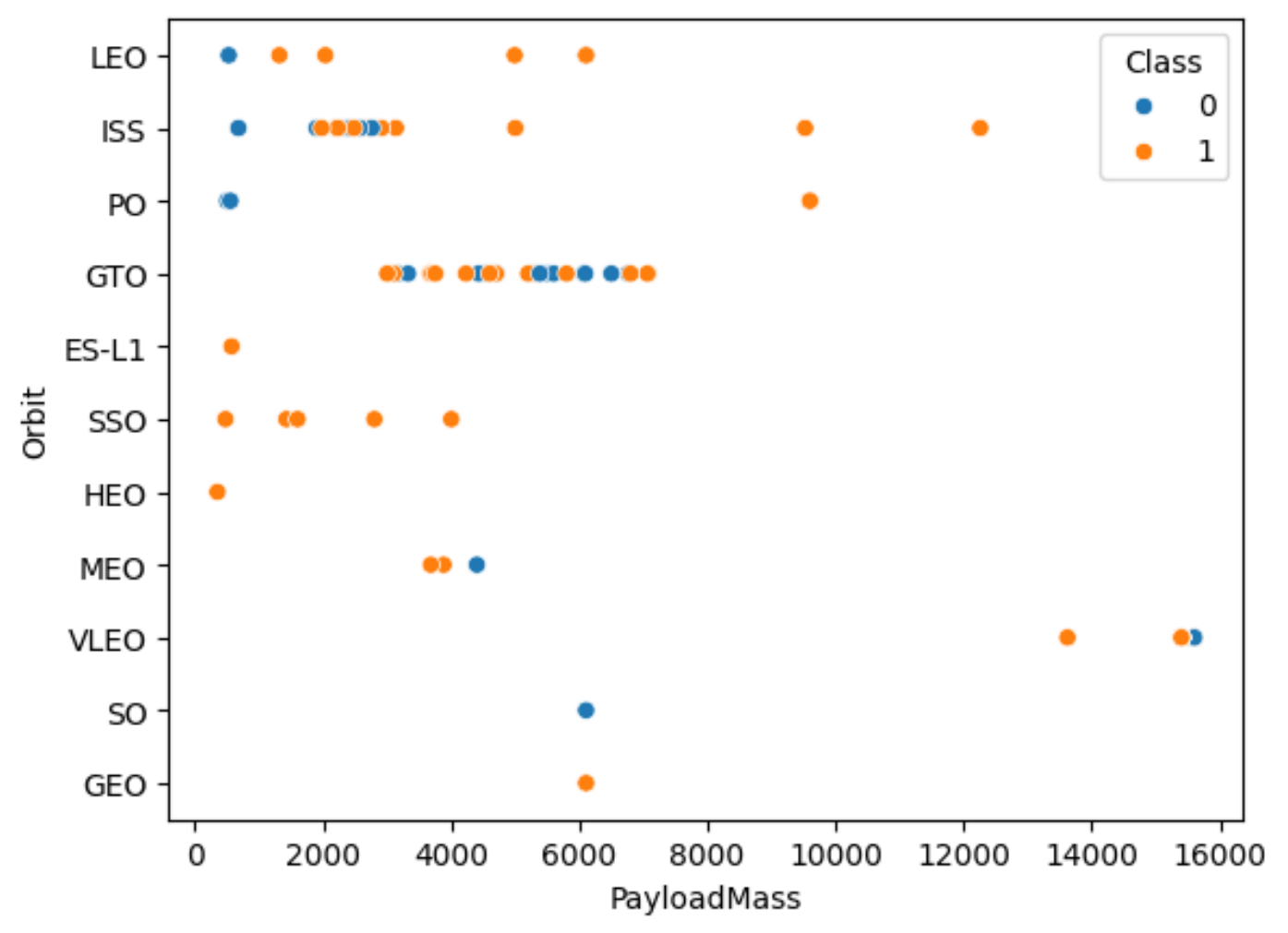
Flight Number vs. Orbit Type

- Scatter plot shows relationship between flight number and orbit type
- Colors represent landing success (1) and failure (0)
- Success rate increases over time across different orbit types
- Later missions show more successful landings



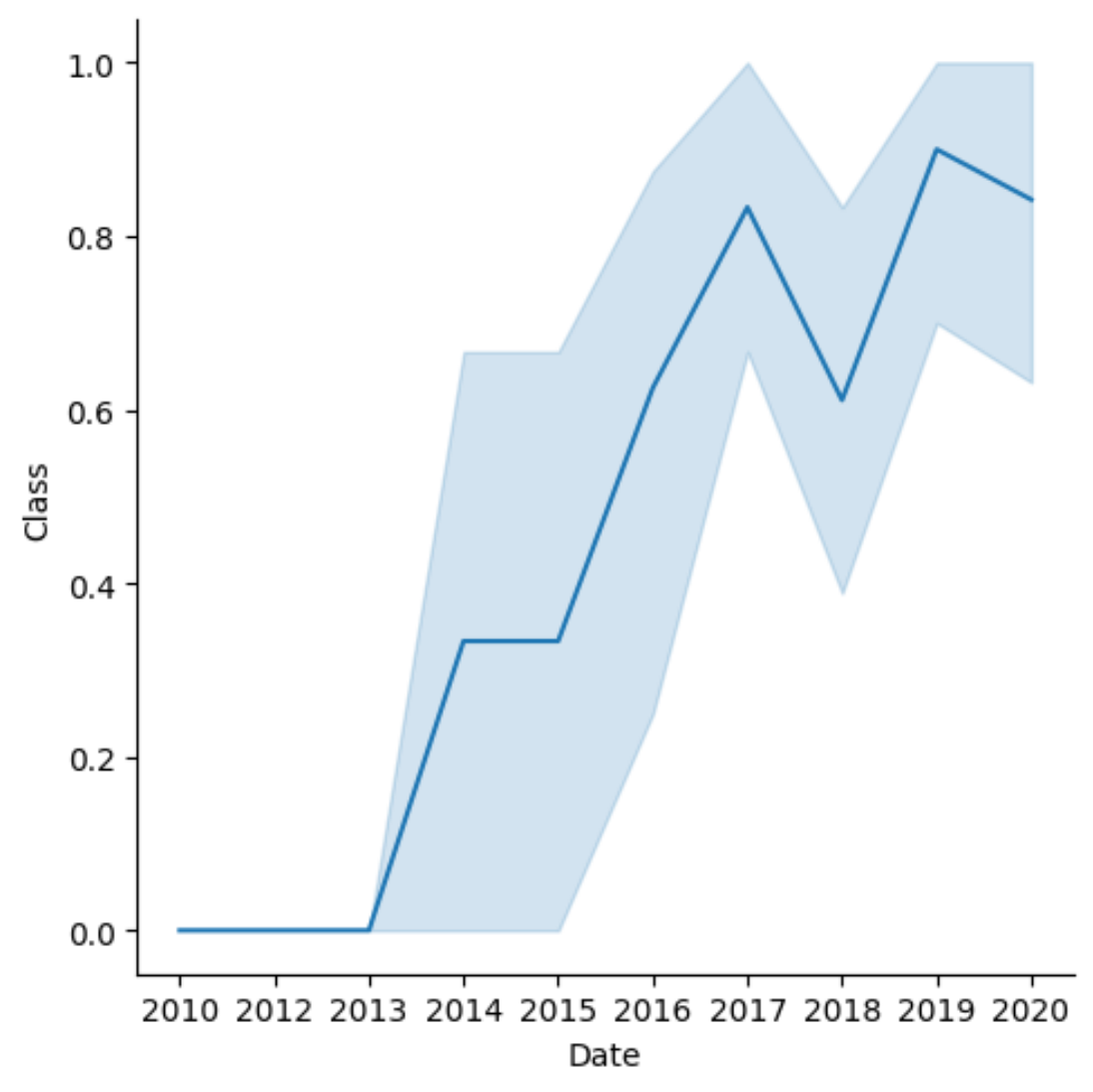
Payload vs. Orbit Type

- Scatter plot shows relationship between payload mass and orbit type
- Colors represent landing success (1) and failure (0)
- Lower payloads are associated with higher success rates
- More complex orbits tend to have higher payloads and lower success



Launch Success Yearly Trend

- Line chart shows yearly average landing success rate
- Success rate increases steadily over time
- Indicates improvement in technology and operational experience
- Demonstrates effectiveness of rocket reuse strategy



All Launch Site Names

- These sites are used for different mission types.

```
%sql SELECT DISTINCT(Launch_Site) FROM "SPACEXTBL";
```

```
 * sqlite:///my_data1.db
Done.
```

Launch_Site
CCAFS LC-40
VAFB SLC-4E
KSC LC-39A
CCAFS SLC-40

Launch Site Names Begin with 'CCA'

```
%sql SELECT * FROM "SPACEXTBL" WHERE "Launch_Site" LIKE 'CCA%' LIMIT 5;
```

```
 * sqlite:///my_data1.db
Done.
```

Date	Time (UTC)	Booster_Version	Launch_Site	Payload	PAYLOAD_MASS__KG_	Orbit	Customer	Mission_Outcome	Landing_Outcome
2010-06-04	18:45:00	F9 v1.0 B0003	CCAFS LC-40	Dragon Spacecraft Qualification Unit	0	LEO	SpaceX	Success	Failure (parachute)
2010-12-08	15:43:00	F9 v1.0 B0004	CCAFS LC-40	Dragon demo flight C1, two CubeSats, barrel of Brouere cheese	0	LEO (ISS)	NASA (COTS) NRO	Success	Failure (parachute)
2012-05-22	7:44:00	F9 v1.0 B0005	CCAFS LC-40	Dragon demo flight C2	525	LEO (ISS)	NASA (COTS)	Success	No attempt
2012-10-08	0:35:00	F9 v1.0 B0006	CCAFS LC-40	SpaceX CRS-1	500	LEO (ISS)	NASA (CRS)	Success	No attempt
2013-03-01	15:10:00	F9 v1.0 B0007	CCAFS LC-40	SpaceX CRS-2	677	LEO (ISS)	NASA (CRS)	Success	No attempt

Total Payload Mass

- The total payload carried by boosters from NASA is 99980 Kg.

```
%sql SELECT SUM(PAYLOAD_MASS__KG_) FROM "SPACEXTBL" WHERE "Customer" LIKE 'NASA%' ;
```

```
 * sqlite:///my_data1.db
Done.
```

SUM(PAYLOAD_MASS__KG_)
99980

Average Payload Mass by F9 v1.1

- Taverage payload mass carried by booster version F9 v1.1 is 2534.67 Kg.

```
%sql SELECT AVG(PAYLOAD_MASS__KG_) FROM "SPACEXTBL" WHERE "Booster_Version" LIKE 'F9 v1.1%' ;
```

```
 * sqlite:///my_data1.db
Done.
```

AVG(PAYLOAD_MASS__KG_)
2534.6666666666665

First Successful Ground Landing Date

- The first successful landing outcome on ground pad was in 22 of July of 2018.

```
%sql SELECT MIN(Date) FROM "SPACEXTBL" WHERE "Landing_Outcome" = 'Success' ;
```

```
 * sqlite:///my_data1.db
Done.
```

MIN(Date)
2018-07-22

Successful Drone Ship Landing with Payload between 4000 and 6000

- The names of boosters which have successfully landed on drone ship and had payload mass greater than 4000 but less than 6000

```
%sql SELECT Booster_Version FROM "SPACEXTBL" WHERE "Landing_Outcome" = 'Success' AND "PAYLOAD_MASS__KG_" > 4000 AND "PAYLOAD_MASS__KG_" < 6000 ;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version
F9 B5 B1046.2
F9 B5 B1047.2
F9 B5 B1048.3
F9 B5 B1051.2
F9 B5B1060.1
F9 B5 B1058.2
F9 B5B1062.1

Total Number of Successful and Failure Mission Outcomes

- Total number of successful and failure mission outcomes

```
%sql SELECT Mission_Outcome, COUNT(*) FROM "SPACEXTBL" GROUP BY "Mission_Outcome";
```

```
 * sqlite:///my_data1.db
Done.
```

Mission_Outcome	COUNT(*)
Failure (in flight)	1
Success	98
Success	1
Success (payload status unclear)	1

Boosters Carried Maximum Payload

- The names of the booster which have carried the maximum payload mass

```
%sql SELECT Booster_Version, PAYLOAD_MASS__KG_ FROM "SPACEXTBL" WHERE "PAYLOAD_MASS__KG_" = (SELECT MAX(PAYLOAD_MASS__KG_) FROM "SPACEXTBL") ;
```

```
 * sqlite:///my_data1.db
Done.
```

Booster_Version	PAYLOAD_MASS__KG_
F9 B5 B1048.4	15600
F9 B5 B1049.4	15600
F9 B5 B1051.3	15600
F9 B5 B1056.4	15600
F9 B5 B1048.5	15600
F9 B5 B1051.4	15600
F9 B5 B1049.5	15600
F9 B5 B1060.2	15600
F9 B5 B1058.3	15600
F9 B5 B1051.6	15600
F9 B5 B1060.3	15600
F9 B5 B1049.7	15600

2015 Launch Records

- The failed landing_outcomes in drone ship, their booster versions, and launch site names for in year 2015

Month_Name	Landing_Outcome	Booster_Version	Launch_Site
January	Failure (drone ship)	F9 v1.1 B1012	CCAFS LC-40
February	Controlled (ocean)	F9 v1.1 B1013	CCAFS LC-40
March	No attempt	F9 v1.1 B1014	CCAFS LC-40
April	Failure (drone ship)	F9 v1.1 B1015	CCAFS LC-40
April	No attempt	F9 v1.1 B1016	CCAFS LC-40
June	Precluded (drone ship)	F9 v1.1 B1018	CCAFS LC-40
December	Success (ground pad)	F9 FT B1019	CCAFS LC-40

Rank Landing Outcomes Between 2010-06-04 and 2017-03-20

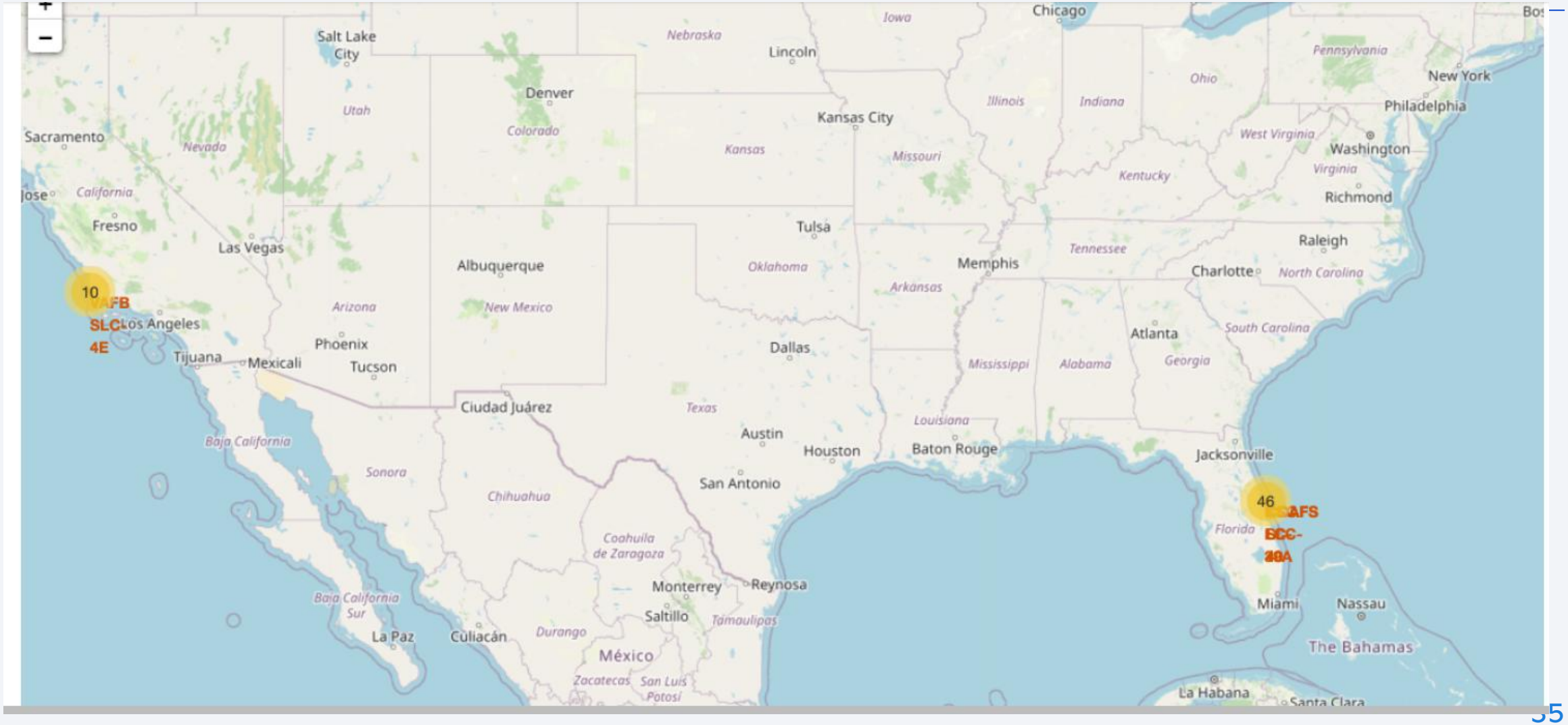
- Rank the count of landing outcomes (such as Failure (drone ship) or Success (ground pad)) between the date 2010-06-04 and 2017-03-20, in descending order

Landing_Outcome	Count
No attempt	10
Success (drone ship)	5
Failure (drone ship)	5
Success (ground pad)	3
Controlled (ocean)	3
Uncontrolled (ocean)	2
Failure (parachute)	2
Precluded (drone ship)	1

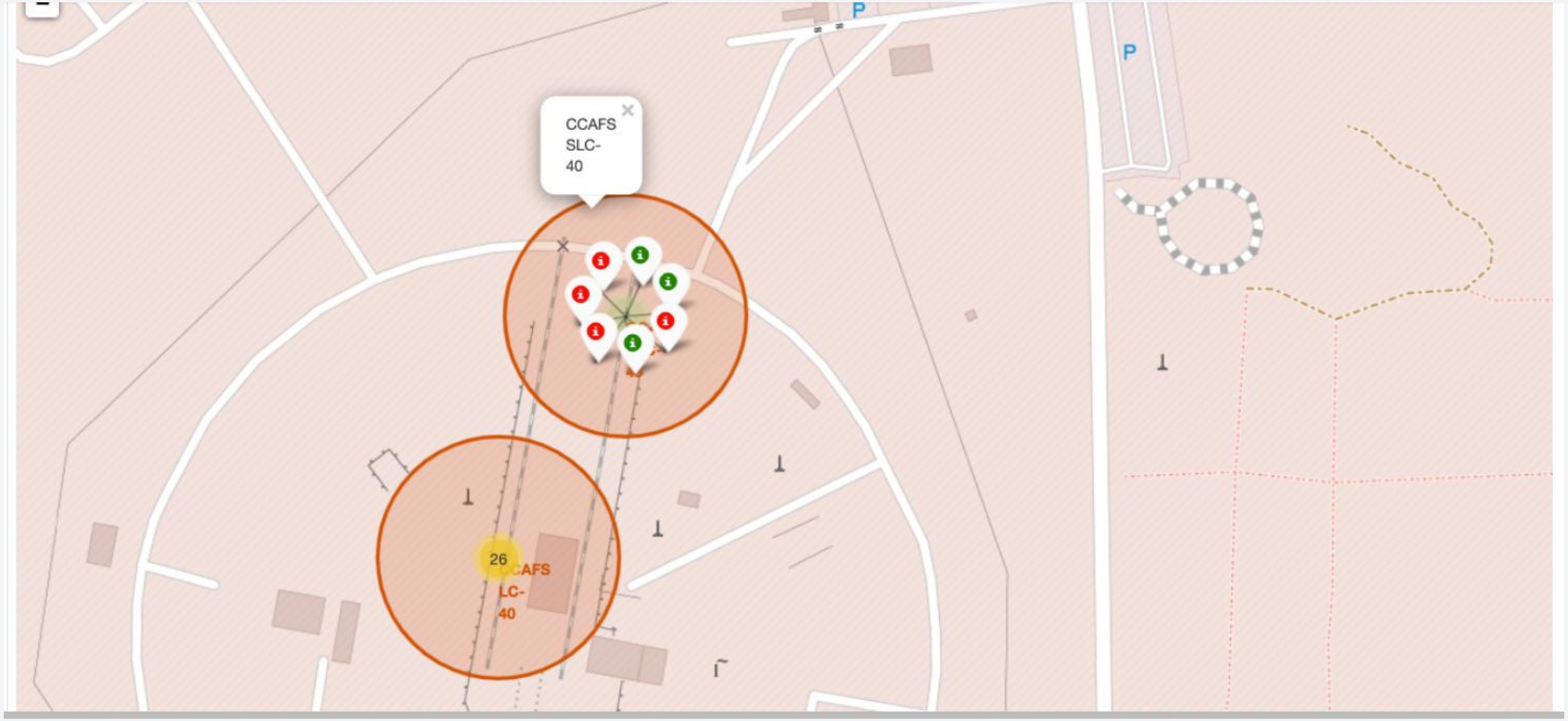
Section 3

Launch Sites Proximities Analysis

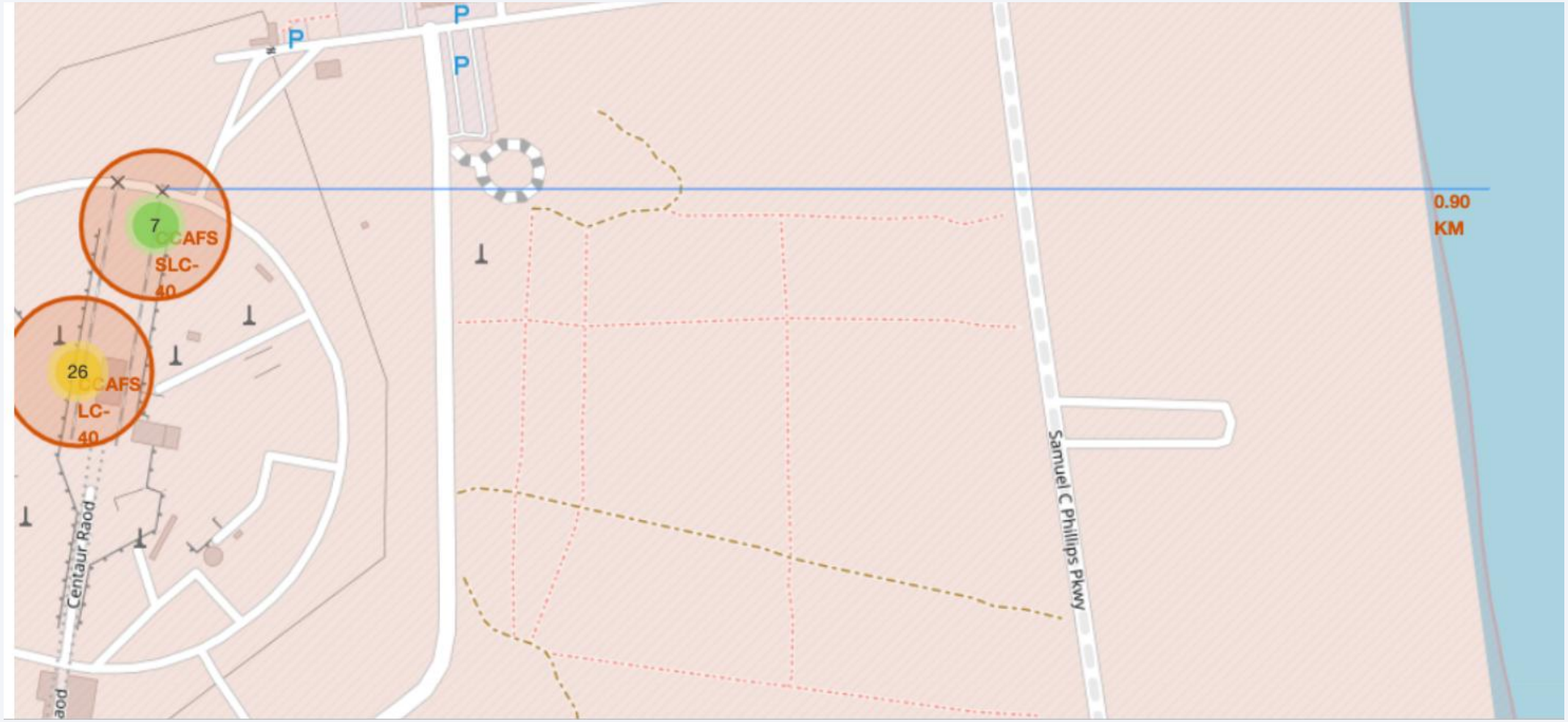
SpaceX Launch Sites Map



Launch Outcomes by Location



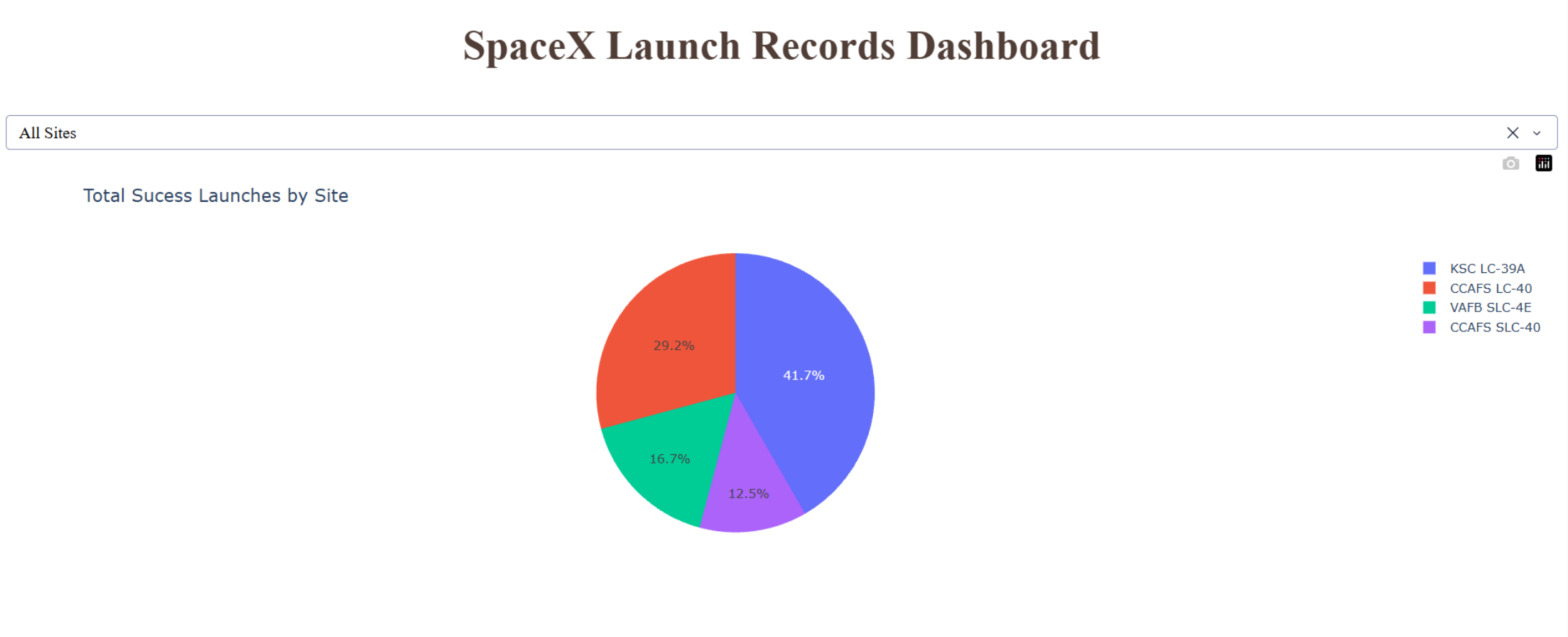
Launch site Proximity Analysis



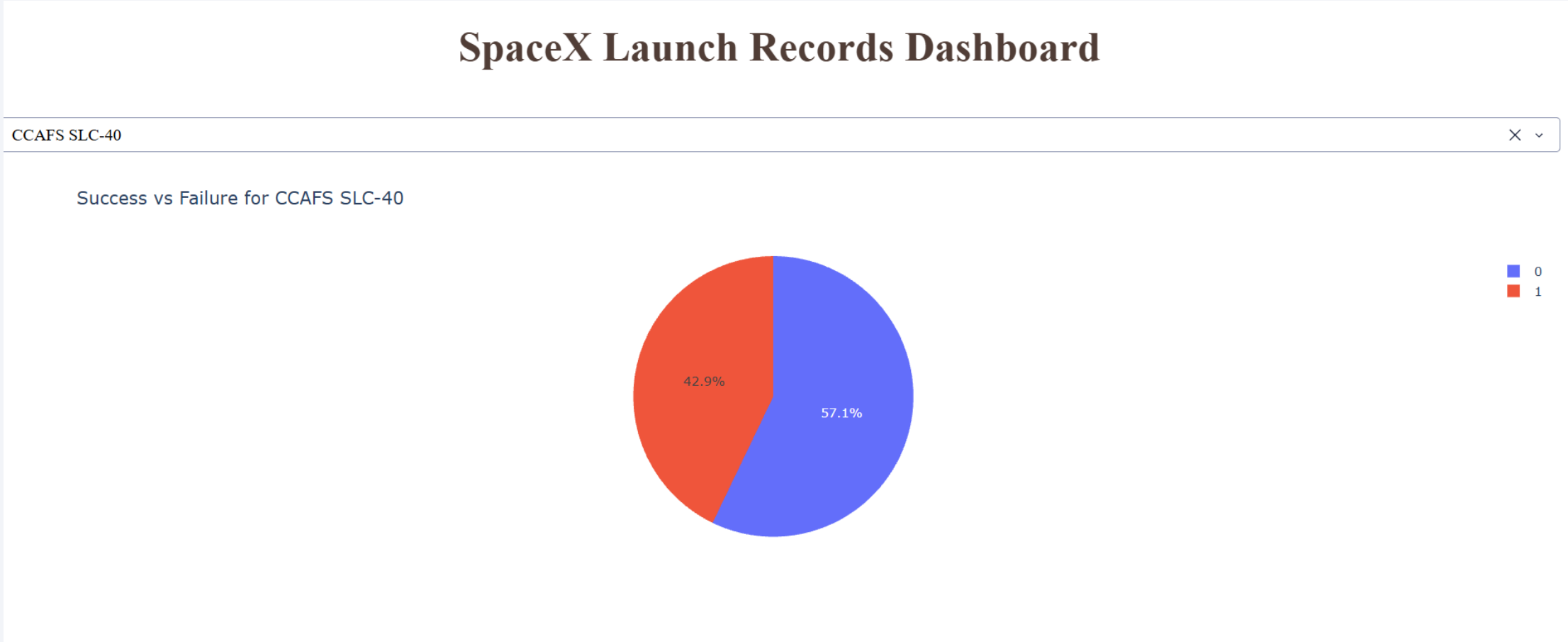
Section 4

Build a Dashboard with Plotly Dash

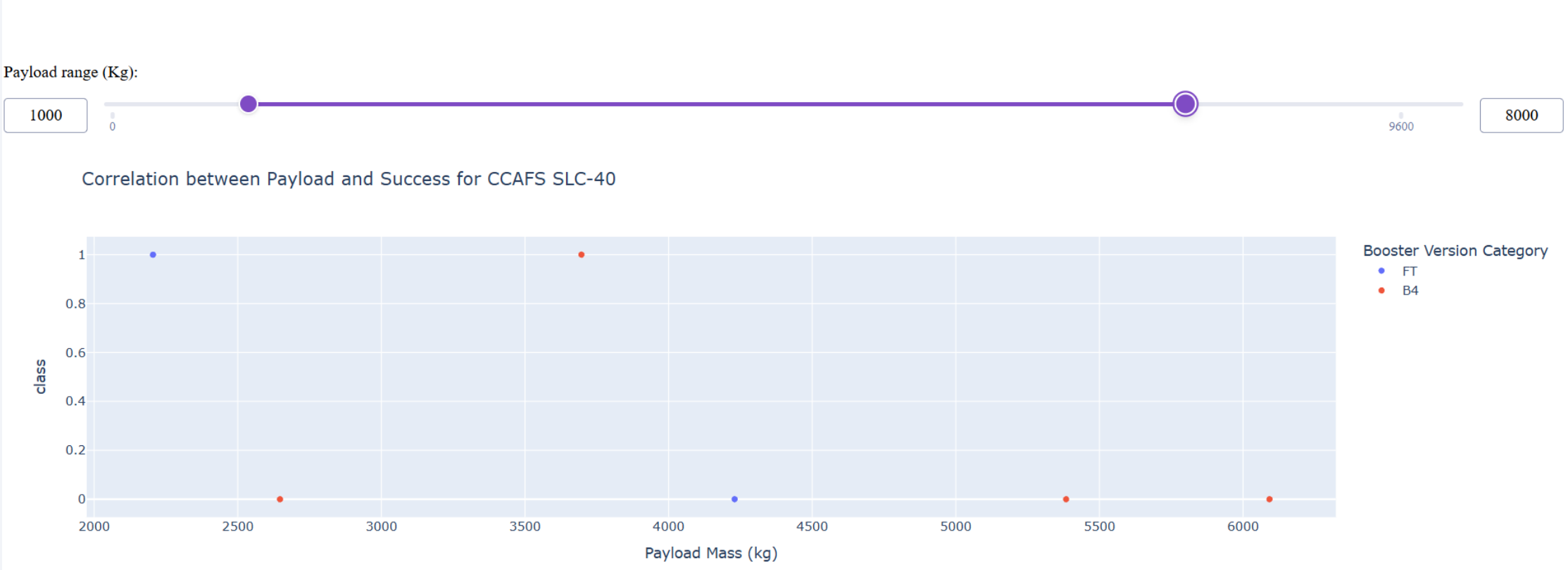
Launch Success Distribution by Site



Launch Success for Top Performing Site



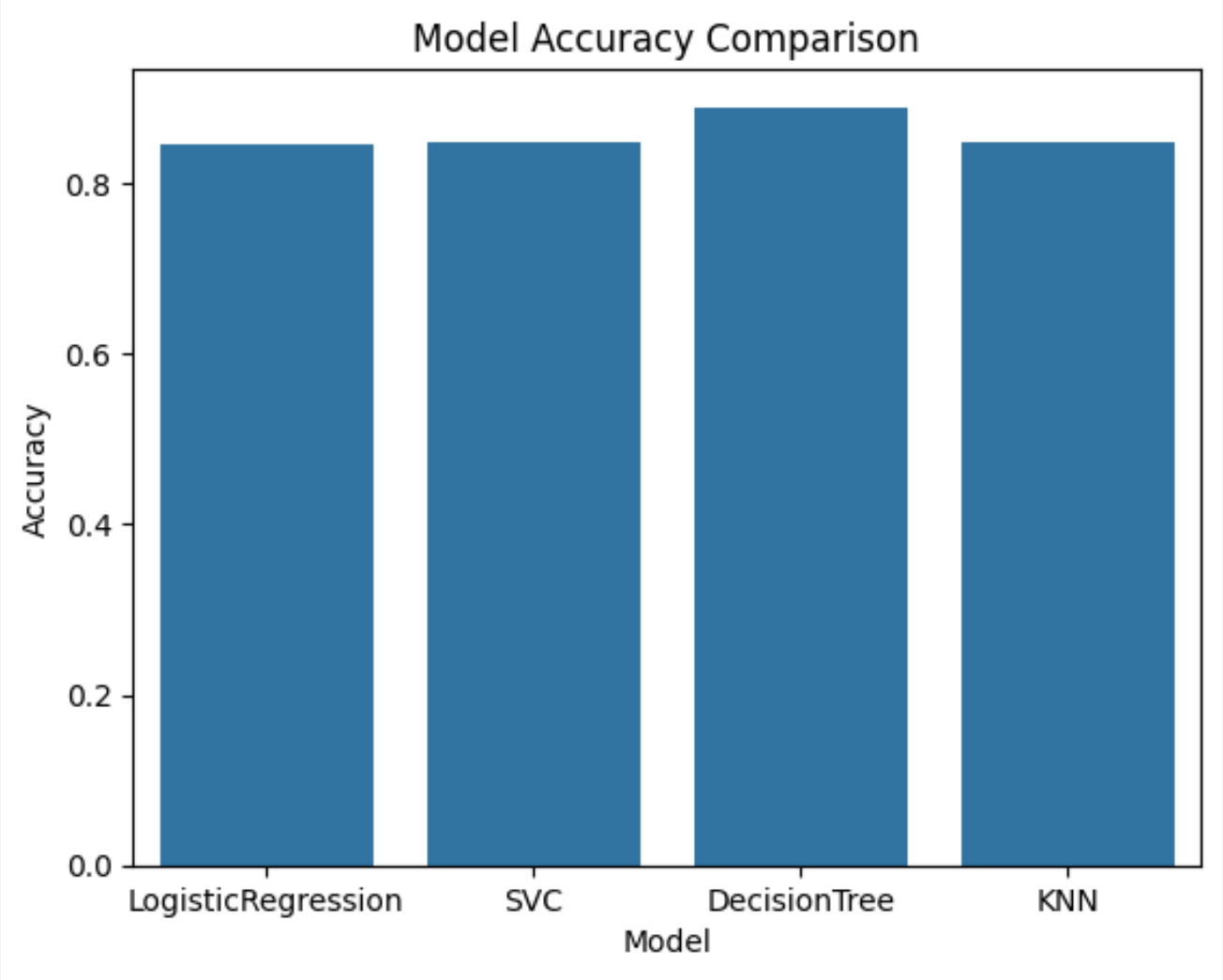
Payload vs Launch Outcome Analysis



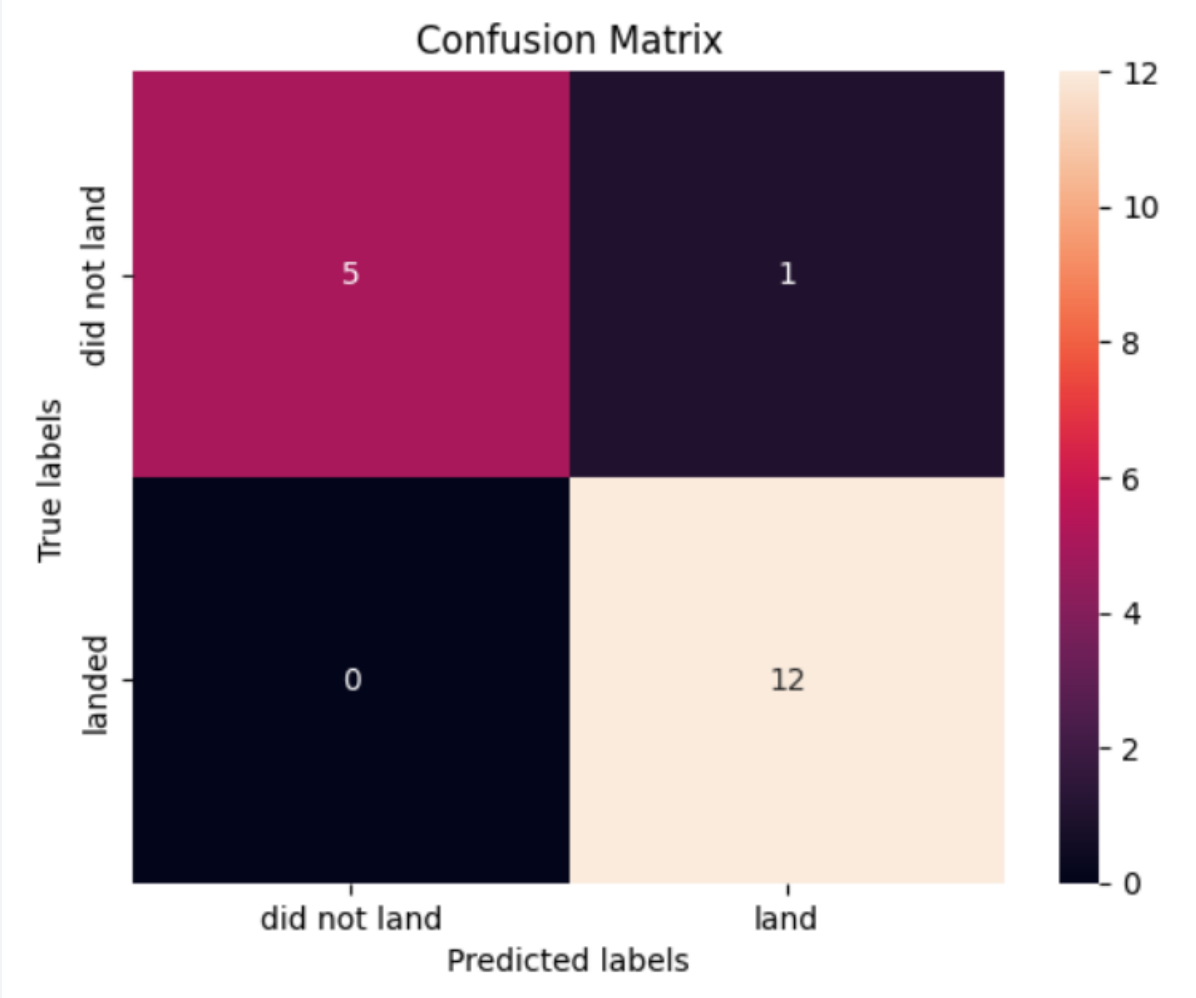
Section 5

Predictive Analysis (Classification)

Classification Accuracy



Confusion Matrix



Conclusions

- Launch success is influenced by launch site and payload mass.
- Some launch sites show higher success rates, especially CCAFS LC-40.
- There is a relationship between payload and success outcome (mid-range payloads tend to perform better).
- The Decision Tree model achieved the highest accuracy (~0.89) compared to other models.
- Data visualization and dashboards help to identify patterns and improve prediction performance.

Thank you!